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The Price of Winning: Analyzing Salary Strategy in the NBA

Introduction

It is commonly thought that professional sports teams, especially in the NBA, that spend more money tend to have more overall success. However, there have been expensive teams that struggle and teams filled with superstars that don't meet the hype and crumble in the playoffs, putting this claim in a new light. Our objective with this project is to investigate to which extent this theory is true.

Our main question is this: **how are modern NBA teams turning salary into success?** Within this topic, we evaluate the following:

- Do higher payroll teams win more? What are characteristics of expensive and cheaper teams?
- Playoff success: do expensive teams perform better in the playoffs?
- At the player level, which players (or player archetypes) are getting paid more? Does this work?
- Is superstar spending more efficient than a balanced roster?
- Which teams win more per dollar? What other patterns emerge when looking at cost per win?

Money management in the NBA can turn out to be the difference between an organization playing their cards right, winning it all and potentially becoming a dynasty and an organization jeopardizing star players and future success. Being informed on prior trends and patterns is crucial for smart decision-making within an organization. This project also applies our acquired skills throughout our time in CS216 and requires that we use them in a variety of ways to analyze a real-world topic.

Data Sources

NBA Team and Player Salaries Datasets (2015-16 through 2025-2026 seasons):

NBA Team Salaries [HoopsHype];NBA Player Salaries [ESPN]

These datasets include information on NBA players and teams' annual salaries from the 2015-16 through 2025-2026 seasons. These include teams' names and salaries and players' names, teams and salaries.

NBA Player Performance Statistics Datasets (2015-16 through 2025-2026 seasons):

NBA Player Per Game Statistics [Basketball Reference]

This dataset includes NBA regular season and playoff player statistics from the 2015-16 through 2025-2026 seasons. These include performance metrics such as per-game averages and efficiency ratings.

NBA Team Performance Statistics Datasets (2015-2016 through 2025-2026 seasons):

NBA Regular Season Team Stats [Basketball Reference];NBA Playoff Team Stats [Basketball Reference]

This dataset includes information about the regular season and playoff statistics of teams from the 2015-2016 through 2025-2026 seasons. Information includes the conferences in which teams belong to, rankings, win/loss percentages, offensive/defensive ratings and more.

Combining these datasets has allowed for full analysis of team and player-level salary and performance metrics. We used the read_html() function from the pandas module to pull tables from each dataset's website URL. We then cleaned up the data, standardizing column names across data sets to allow for merging on certain columns. For example, we made sure the 'Team' column stays consistent throughout all data sets, only identifying teams by their mascot names. We additionally added a 'Season' column throughout all data sets to allow for proper analysis of data across seasons. We then created new statistics (TopHeavyIndex, TeamType, etc.) and used them to examine patterns and formulate relationships.

Modules

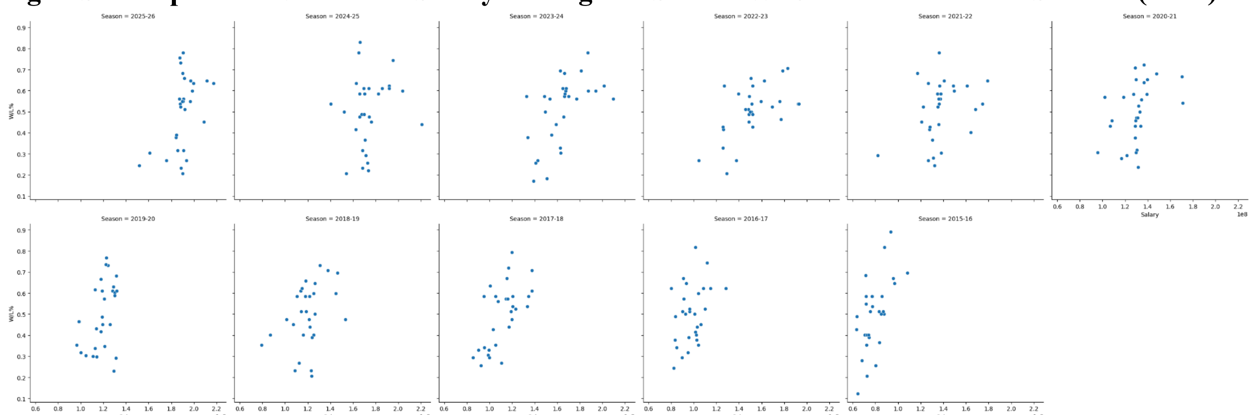
Module 3: Visualization

Visualization played a primary role in communicating the patterns between salary and success. By creating multiple visualizations, we examined several relationships such as team salaries vs. regular season win rate, team salaries vs. playoff success, performance vs. salary distributions, and more. Scatter plots, bar plots, as well as other various relational and categorical plots are used to represent the relationships noted above. We used module 3 during the data investigation, analysis and final report stages of our project.

Module 4: Data Wrangling
We applied the Data Wrangling module to clean and format our salary and performance datasets. These datasets come from different sources and are not guaranteed to be in consistent, Python-ready formats, so they must be transformed into a unified structure before analysis. Using techniques from this module, we imported the datasets using pandas, inspected their structure, cleaned string columns using vectorized string operations, and converted columns to proper numeric types. We used module 4 primarily during the data gathering and cleaning stages of our project.

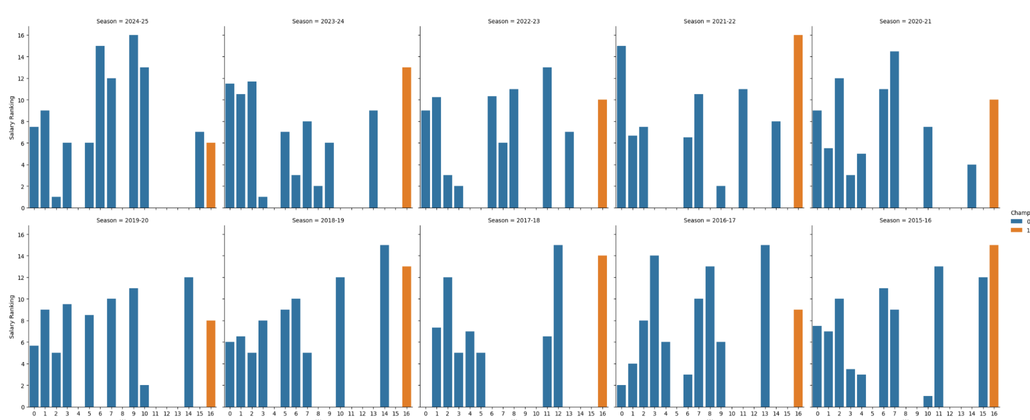
Module 6: Combining Data
This module allowed us to merge salary datasets with team performance statistics and advanced metrics across seasons. We combined data to calculate efficiency metrics such as cost per win and evaluate whether spending correlates with success factors like wins and losses. Module 6 has been primarily used during the data cleaning and investigating stages of our project.

Results & Methods (found on https://duke.box.com/s/n6nm8qkgcd9habvrdpouum0mtndt6ccu)
Fig 1: Scatterplots of NBA Team Salary vs. Regular Season Win/Loss Ratio across Seasons (desc.)



Team salaries league wide have increased substantially over the past decade, as we see the clusters of points traversing rightward on the x-axis as the years increase. Additionally, though the strength varies season-to-season, there seems to be a positive correlation between salary and W/L ratio, signifying that higher payroll teams win more during the regular season. However, when you look at the playoffs…

Fig 2: NBA Team Playoff Wins (W) vs. Playoff Team Salary Rankings across Seasons (desc.)
16 "W" means that a team won all 4 best-of-7 series and won the NBA Championship (hued in orange)



...only 2 of the past 10 NBA Champions had the highest salaries of all teams in the playoffs. We also see that lower payroll teams (relative to others in the playoffs) tend to not fare as well.

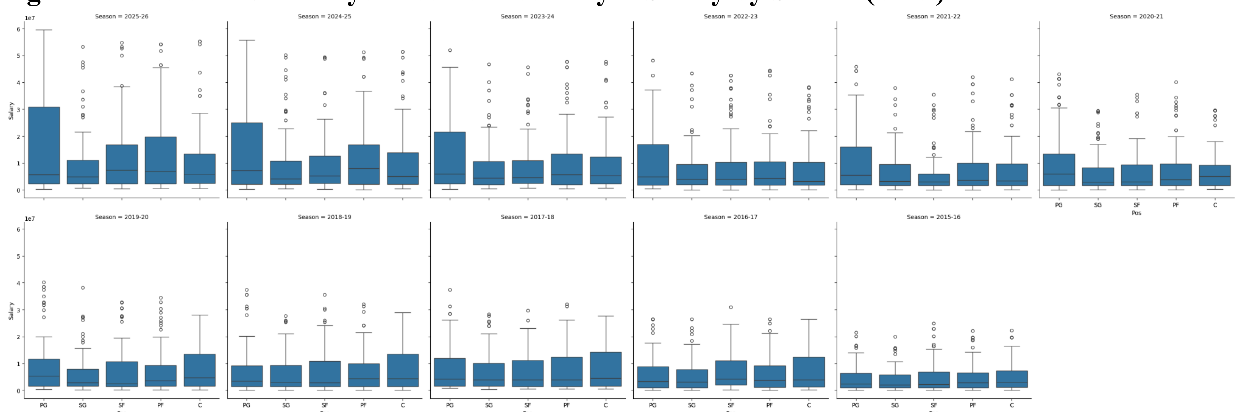
Incorporating data at the player level, we can look closer at how these teams are spending their money. The median NBA player was making $2,547,210 in the 2015 season and now makes $5,742,480 a decade later, confirming the league-wide increase in pay observed in **Fig 1**. Positionally, the highest paid player (year-to-year) in the league since the 2017-18 season has been a point guard (Stephen Curry, GSW). When we look at trends over time:

Fig 3: Pivot Table of Median NBA Player Salaries, Indexed by Position, Across Seasons

Year	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26
Pos											
PG	2399040.0	3386598.0	4221000.0	3437580.0	5327563.5	6000000.0	5495532.0	4833600.0	5999455.5	7152094.5	5685000.0
SG	2148360.0	3101940.0	4001740.0	3078840.0	2803357.0	2816760.0	3261480.0	4000000.0	4423440.0	4041249.0	4831558.5
SF	2288205.0	4264057.0	3853931.0	2791440.0	2475840.0	2954700.0	3053760.0	3987119.0	4556983.0	5200000.0	7278846.0
PF	2825093.0	3800000.0	3974999.5	4320500.0	3576277.5	3837260.0	3676852.0	4325100.5	5709877.0	8000000.0	6914260.0
C	2943221.0	4000000.0	4593799.5	4350000.0	4767000.0	5000000.0	3277080.0	3200000.0	5302096.0	5007280.0	5850540.0

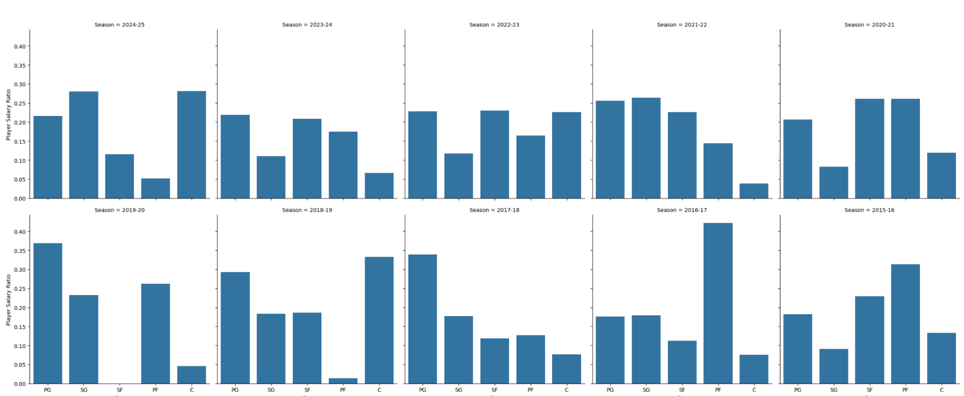
Forwards (SF, PF) and centers (C) had higher median earnings until around 2019, and since then point guards (PG) and forwards dominated the median earnings, reflecting a league-wide investment in these positions. Visualizing this relationship:

Fig 4: Box Plots of NBA Player Positions vs. Player Salary by Season (desc.)



We observe the surge in point guard spending over the last decade, specifically in higher quartiles. While forwards and centers had higher median/upper quartile earnings than other positions until 2020, these earnings have been higher for point guards and forwards ever since, solidifying our observation from **Fig 3**. There are outliers present in nearly every position throughout each season, and we can assume that those are the "superstars" of the league, warranting an outstanding pay. To see if these changes in position pay have affected team success, we look at positional pay distributions of NBA Championship teams.

Fig 5: Salary Ratios by Position of NBA Championship Teams across Seasons (desc.)



We see that some teams had guard-heavy, some forward-heavy, and others more balanced pay structures. It seems that there is not one positional pay structure that guarantees championships, nor is one the norm.

From another angle, we dive further into superstar spending. We constructed a metric to quantify how concentrated a team's salary is among its highest-paid players, `TopHeavyIndex`. This is the proportion of a team's total payroll that is allocated to its top three highest-paid players:

$$\text{TopHeavyIndex} = \frac{\text{Salary of Top 3 Players}}{\text{Total Team Salary}}$$

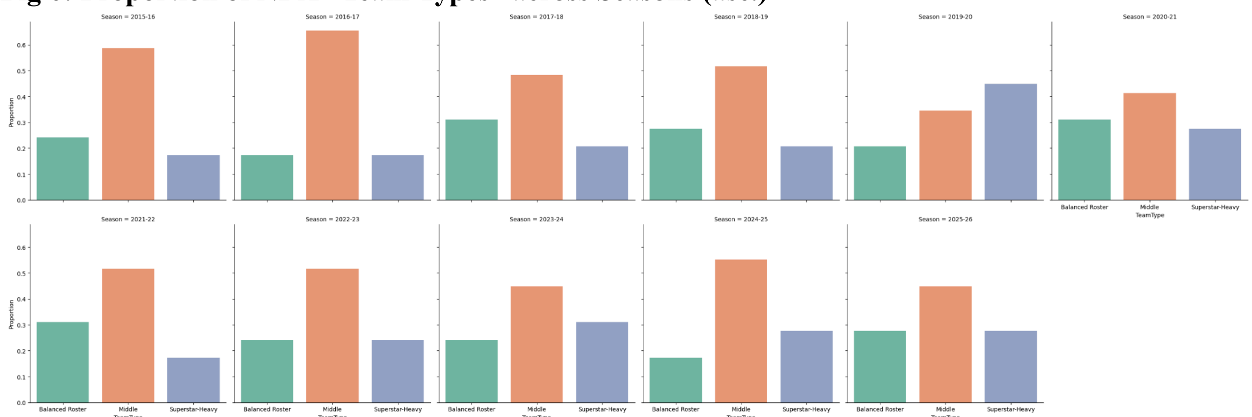
A TopHeavyIndex value of 0.75 indicates that 75% of the team's total salary is allocated to its top 3 players.

Using this metric, we categorized each team (per season) into one of three groups:

- **Balanced Roster:** teams in the bottom quartile (most evenly distributed payroll)
- **Middle:** teams in between
- **Superstar-Heavy:** teams in the top quartile of TopHeavyIndex values (teams with the highest concentration of team salary allocated to its top 3 players)

This classification is based purely on salary concentration rather than subjective labeling of players, allowing for a consistent, data-driven definition of "superstar-heavy" teams. To better understand how teams allocate salary across players, we first examine the distribution of team types.

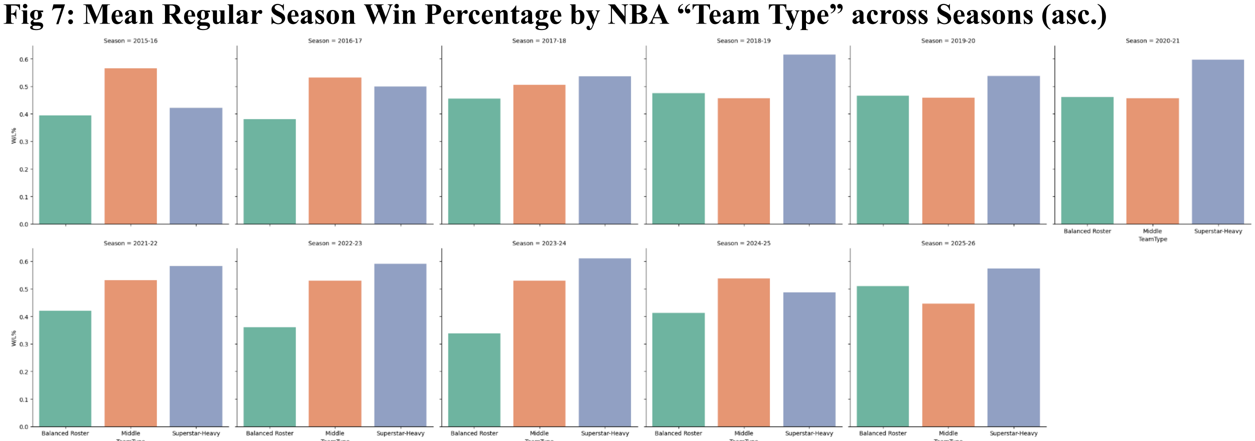
Fig 6: Proportion of NBA "Team Types" across Seasons (asc.)



From a visual inspection of the data, it is clear that the majority of NBA teams fall into the "Middle" category. However, the gap has decreased over the past decade as we see a rise of both superstar-heavy and balanced rosters.

While these patterns provide insight into how teams structure their payrolls, they do not yet indicate how these strategies impact performance outcomes. To evaluate this, we compare average (regular season) win percentage across team types.

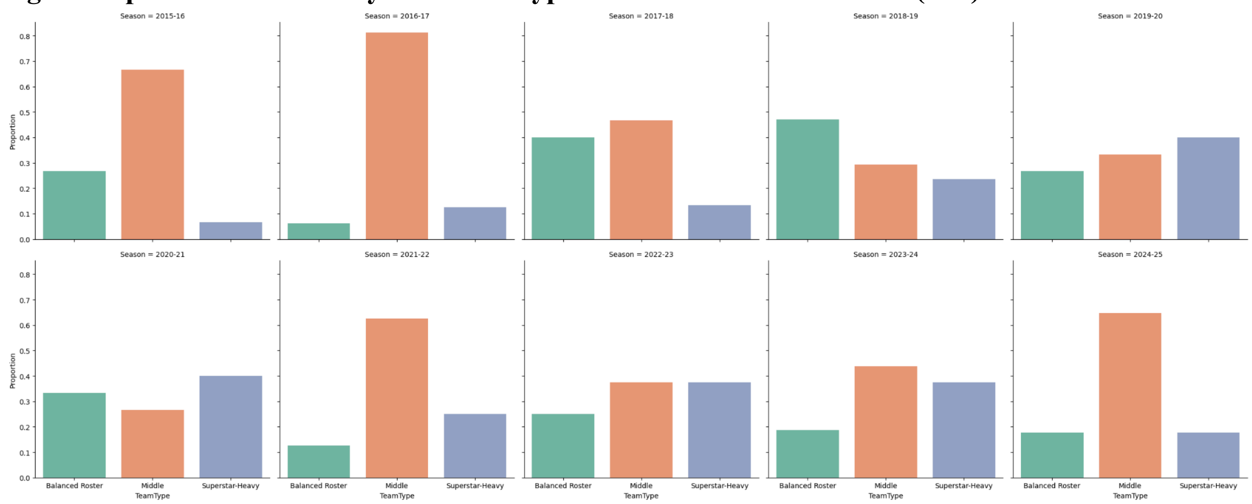
Fig 7: Mean Regular Season Win Percentage by NBA "Team Type" across Seasons (asc.)



From this visualization, we see that since the 2017-18 season superstar-heavy teams garner the highest regular season win percentage across the NBA. Additionally, in 7 out of the 10 last seasons, the balanced rosters have the lowest win percentage of the three types of teams.

Though the differences in win percentage across team types might not seem too significant at times, these margins can decide whether a team makes the playoffs or not.

Fig 8: Proportion of NBA Playoff "Team Types" across Previous Seasons (asc.)



While "Middle" teams make up the majority of 6 of the last 10 playoff team types, the proportion of superstar-heavy and balanced rosters in the playoffs have been pretty variable.

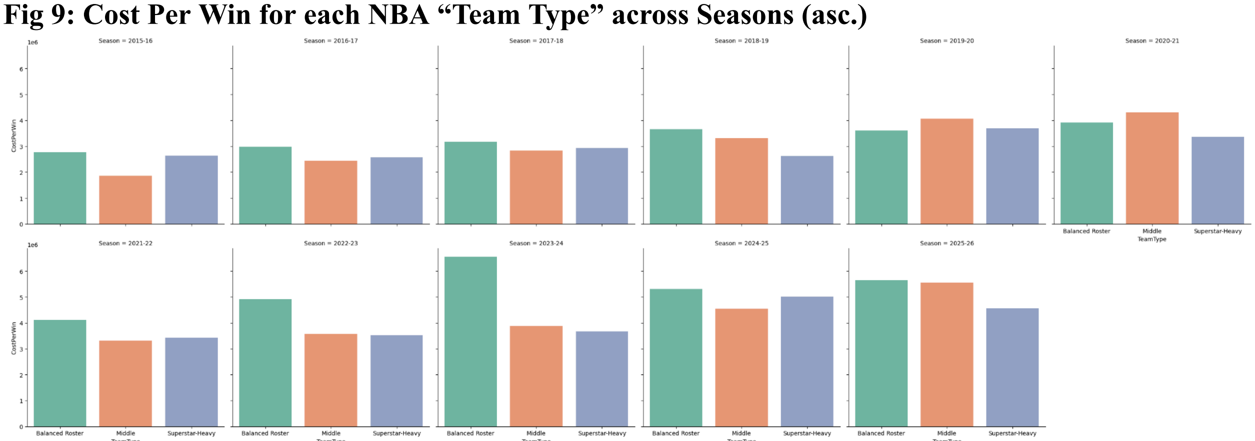
Looking at numbers again, of the past 10 NBA Champions, 7 have been of the "Middle" type and the other 3 have been superstar-heavy (2020 Lakers, 2021 Bucks, 2022 Warriors).

To analyze team efficiency aside from just wins/losses, we introduce `cost per win`, defined as:

$$\text{Cost Per Win} = \frac{\text{Total Team Salary}}{\text{Number of Wins}}$$

This metric aims to represent the amount of money a team spends to generate a single regular season win, allowing us to evaluate financial efficiency rather than just performance.

Fig 9: Cost Per Win for each NBA "Team Type" across Seasons (asc.)



There is a substantial increase in cost per win over the years, reflecting the league-wide increase in salaries observed earlier. Since 2015, it seems that balanced rosters have been warranting a higher cost per win than other types of teams, with superstar-heavy teams staying relatively lower in the same stat.

Limitations & Future Work

The majority of our limitations presented themselves during the data gathering phase of our project. For one, there were not many NBA player salary datasets out there. The only one that was not behind a paywall did not contain salary data for certain players for no identifiable reasons, but we had no other option but to use it. While general patterns were still feasible to analyze (we had performance/salary data of 4600+ players), this caveat could call the more specific observations we made into question (e.g. our superstar queries, which focused on a small subset of outlying salaries). This might not have biased the data in any obvious direction, since the missing values did not appear to follow a clear pattern, but it still proved cautionary for us.

In terms of future work, one could implement some heavier statistical and predictive methods into our work to determine whether the patterns we observed are statistically robust. Our aim was for an exploratory analysis of data but this has a lot of potential for more developed breakdowns. Observed patterns could also be clearer and more credible if the data we were missing is filled in.

Conclusion

Over the past decade, NBA salaries have increased drastically. Our analysis shows that spending does matter, but not in the sense of simply “buying wins”. Across seasons, higher-payroll teams generally posted higher regular season win percentages, suggesting that financial investment does help teams build competitive rosters. However, our playoff analysis shows that the most expensive team does not often win the championship. Together, these findings suggest that payroll is better understood as a tool for contention rather than a guarantee of success.

Our player and team-level results reinforce this idea by showing that allocation matters just as much as total spending. Salaries have increasingly shifted toward point guards and versatile forwards, reflecting the modern NBA’s emphasis on playmaking, perimeter scoring, and multi-dimensional position play. Additionally, teams with “superstar-heavy” payrolls produced stronger regular season results and lower cost per win than more balanced pay structures, while most NBA champions as of late have fallen in the “Middle” category. Overall, our findings suggest that successful teams are not simply the ones that spend the most, but the ones that spend strategically. There seems to exist a balance between generating the greatest competitive value while still maintaining enough roster balance to sustain postseason success.

Collaboration Reflection & Sharing Plans

Our collaboration ended up being far less structured than we had originally planned. In our initial plan, we outlined weekly check-in meetings, a shared calendar and tracking responsibilities in writing, none of which came into fruition. Regardless of the lack of material representation of it, collaboration was still effective. There was frequent communication between members and a general understanding of who was doing what.

If we did this again, we would have not planned on such a structured collaboration plan, as it seems we underestimated how busy we all would be during the semester. Thus, we would have optimized the collaboration plan to be more flexible and tailored to each of us as individuals, dividing the work up as such, too.

In terms of sharing our project, we plan on pushing everything we have done to a github repository, enabling us to display it on all of our profiles. It is assumed that some group members will be using this project as part of a portfolio/resume of theirs as well, and we are all consenting to this being done.

AI Disclosure

Artificial Intelligence Tools: ChatGPT (OpenAI, GPT-5, February-April 2026)

Conceptualization: ChatGPT was used to help refine the project scope and organize the NBA “Moneyball” topic into specific research questions (payroll vs wins, cost-per-win efficiency, superstar vs balanced roster, and playoff performance), as well as to suggest ways to divide responsibilities among group members.

Information Collection: We used ChatGPT to assist in providing us useful player and team salary and performance data sets to use in our project.

Planning & Methodology: ChatGPT provided guidance on appropriate course modules for the project and suggested analytical approaches such as visualization strategies, dataset merging, and statistical comparisons. ChatGPT also assisted with error handling and instruction on more complex data wrangling methods.